

MatchMaker: A Deep Learning Framework for Drug Synergy Prediction

Halil Ibrahim Kuru^{ID}, Ozgur Tastan^{ID}, and A. Ercument Cicek^{ID}

Abstract—Drug combination therapies have been a viable strategy for the treatment of complex diseases such as cancer due to increased efficacy and reduced side effects. However, experimentally validating all possible combinations for synergistic interaction even with high-throughput screens is intractable due to vast combinatorial search space. Computational techniques can reduce the number of combinations to be evaluated experimentally by prioritizing promising candidates. We present MatchMaker that predicts drug synergy scores using drug chemical structure information and gene expression profiles of cell lines in a deep learning framework. For the first time, our model utilizes the largest known drug combination dataset to date, DrugComb. We compare the performance of MatchMaker with the state-of-the-art models and observe up to $\sim 15\%$ correlation and $\sim 33\%$ mean squared error (MSE) improvements over the next best method. We investigate the cell types and drug pairs that are relatively harder to predict and present novel candidate pairs. MatchMaker is built and available at <https://github.com/tastanlab/matchmaker>

Index Terms—Deep learning, drug synergy, chemical features, cancer cell lines

1 INTRODUCTION

IN complex diseases, multiple cellular mechanisms are often altered in the cell; therefore, treating them with a single drug and focusing on a single target is usually an ineffective strategy. Combination therapy is a promising solution that uses multiple drugs that exhibit better therapeutic efficacy together than the sum of individual drug effects [1]. One other advantage of combination therapy is that by lowering the required dosage per drug, it reduces the risk of adverse effects [2]. Especially, in cancer, where drug resistance is common, combinatorial drug therapy has been an effective strategy for several decades [2], [3], [4]. Thus, pinpointing the right combinations of drugs is an important task that has translational, clinical, and economic impacts.

It is possible to test for synergistic activity of drugs experimentally. However, exhaustive testing of drug combinations is not feasible even with high-throughput screens due to the vast number of drug combinations [5]. Computational methods that predict drug pair synergy can complement and guide high-throughput synergy screens by prioritizing the drug pairs to be further evaluated by experiments [2]. To this end, several algorithms have been proposed to

predict synergistic drug pairs which utilize a diverse set of features such as chemical structure, biological networks interactions (e.g., drug-protein, protein - disease, etc.) and omics data [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. While feature engineering and a systems approach come with the promise of performance increase, features other than the chemical structure are not always available, and biological networks such as protein interaction networks are incomplete. This limits the generalization and applicability of such approaches.

Two algorithms achieve state-of-the-art performances on drug synergy prediction solely by using the drug structures and gene expression profile of the target cell. Preuer *et al.* [18] propose a deep neural network based approach, DeepSynergy, which uses only the chemical fingerprints of the drugs and the gene expression profiles of the cell line of interest. Janizek *et al.* [19] present TreeCombo, an algorithm based on the extreme gradient boosted trees (XGBoost) algorithm [20] on the same dataset with the same set of features and report performance improvements over DeepSynergy.

In this study, we present *MatchMaker*, a deep neural network-based drug synergy prediction algorithm. The model makes use of the drug chemical structures and untreated cell line gene expression profiles to predict the Loewe Additivity score [21] of a drug pair (referred to as Loewe score from now on). Unlike previous work [18], [19], which concatenate the features and predict synergy, MatchMaker trains two parallel subnetworks, one per drug, to learn drug specific representation on a particular cell line. The joint representation for the pair then becomes an input for a third sub-network, which learns to predict drug pair synergy. We show that this approach improves over state-of-the-art performance as learnt embeddings make it easier for the synergy prediction model to predict synergy. Moreover, for the first time in the literature, we utilize the largest drug combination dataset released to this date [22] and benchmark the performances of the state-of-the-art methods on this dataset.

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TABLE 1
DrugComb Dataset Details

	Combinations	Drugs	Cell lines
Initial dataset	466,033	4,150	112
After feature filtering	335,692	3,040	81
After duplicate removal	286,421	3,040	81

MatchMaker achieves 79.49 Mean Squared Error (MSE), 0.79 Pearson and 0.74 Spearman correlation when predicting Loewe scores. This corresponds to 33, 13, and 14 percent improvements over the state of the art, respectively. We investigate the cell lines and drug types for which the model is effective in predicting synergies. We detect novel synergistic pairs that are not marked in the ground truth data set but have reports in the literature as effective pairs. MatchMaker is built and available at <https://github.com/tastanlab/matchmaker>.

2 METHODOLOGY

2.1 Problem Formulation

We model the synergy prediction problem as a regression task. For a given drug pair $\langle i, j \rangle$ and a cell line k , target value is $y_{i,j,k} \in \mathbb{R}$ which is measured by Loewe [21] and positive $y_{i,j,k}$ values indicate synergism. Our dataset contains n unique trios. Each trio contains two drugs i and j represented with x_i and x_j drug feature vectors. The cell line k is represented with c_k cell line feature vector. The problem is to predict $y_{i,j,k}$ by using drug (x_i and x_j) and cell line specific features (c_k).

2.2 Drug Combination Dataset

We use the synergy scores provided in the DrugComb [22] database. The data were obtained from <https://drugcomb.fimm.fi/> (version v1.4, downloaded on Dec 2019). DrugComb curates high-throughput combinatorial drug screening data from different studies such as O’Neil dataset [23], Forcina dataset [24], and NCI Almanac dataset [25].

The dataset contains 466,033 combinations across 112 cell lines and 4,150 drugs. Since we use drug structure and cell line gene expression together as features, we filter this data such that the drugs with the available structure information from PubChem database [27] and cell lines with available

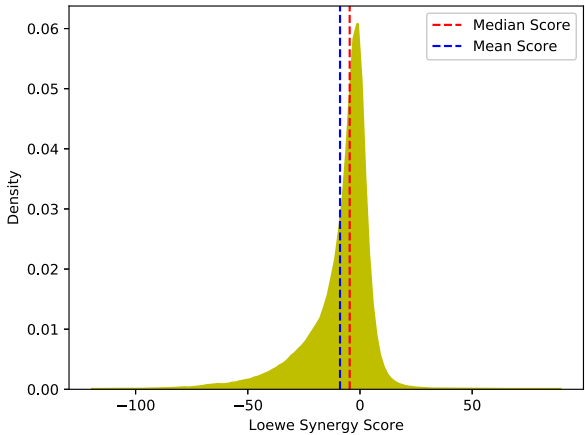


Fig. 1. The distribution of the Loewe scores in the processed DrugComb dataset. The processed data contains 286,421 experiments (drug1, drug2, cell line) for 3,040 drugs, performed on 81 cell lines.

gene expression data from Iorio *et al.* [28]. This leaves us with 335,692 drug pair-cell line combinations that cover 3,040 drugs and 81 cell lines. For the 81 percent of the combinations, there is only a single experiment reporting Loewe score. For the remaining combinations, there are more than one (2 to 8) replicate experiments reporting Loewe scores. For some of these cases, the replicate experiments disagree. That is, some replicates have an equal number of positive Loewe scores (synergism) and negative scores (antagonism) on drug pair, cell line trio. We filter out such 1,861 trios, in which half of the replicates report synergism and other half reports antagonism. For other trios, the Loewe scores of the replicate experiments are averaged. The final processed data contains 286,421 experiments (drug1, drug2, cell line) for 3,040 drugs, performed on 81 cell lines. Table 1 summarizes the number of samples, drugs, and cell lines after the data processing and filtering steps. We should note that not all drug pairs are tested on every cell line, and on average, a drug pair is tested on 25 cell lines. The Loewe distribution of the scores is shown in Fig. 1.

Since DrugComb contains drug combinations from several studies, we further look at the number of combinations for each individual study. Table 2 shows the number of combinations, unique drugs, cell lines and the size of dose-response matrix for each study after data processing steps summarized at Table 1. As illustrated in Table 2, the

TABLE 2
The Data Statistics of the Studies After Preprocessing DrugComb

Study	Number of drug combinations	Number of drugs	Number of cell lines	Size of full dose-response matrix
NCI ALMANAC [25]	266798	101	54	4x4, 4x6
FORCINA [24]	1804	1804	1	2x2
NCATS_2D_3D	10	5	1	10x10
NCATS_ES(FAKI/AURKI)	1834	1835	1	6x6
NCATS_ES(NAMPT +PARP)	44	30	1	6x6, 10x10
NCATS_HL	2385	1829	3	6x6, 10x10
ONEIL [23]	13456	36	28	5x5
YOHE [26]	90	25	1	10x10

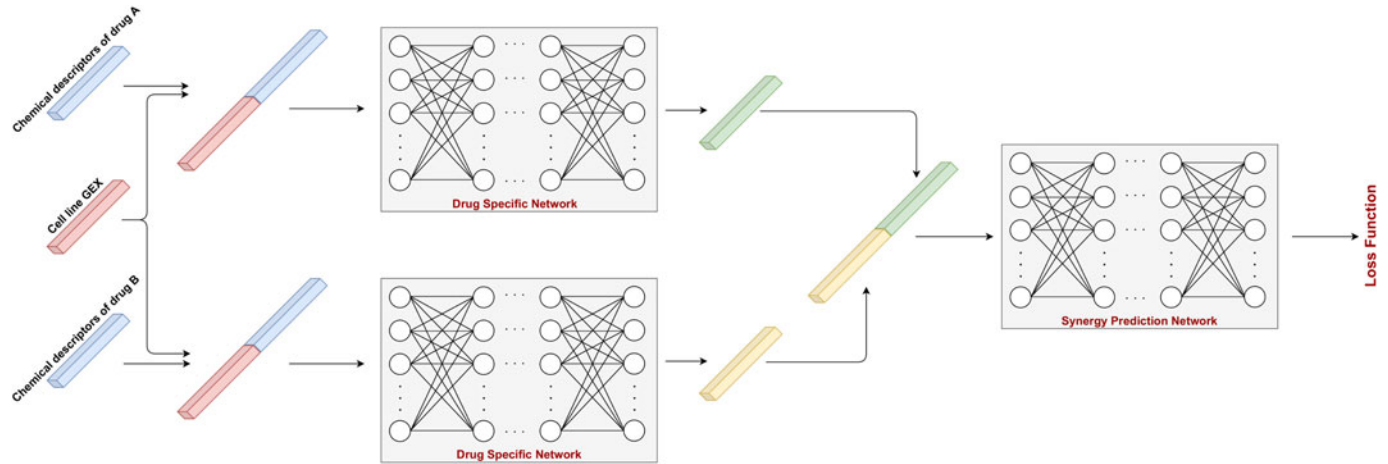


Fig. 2. The architecture of MatchMaker. The chemical features of each drug are concatenated with gene expression features of the cell line and input to drug specific subnetworks (DSNs). The DSNs learn a representation of each drug separately conditioned on cell line gene expressions. Synergy Prediction Network (SPN) takes the representations of both drugs and applies fully connected layers to predict Loewe values. The model is trained to minimize a weighted version of the MSE between actual and predicted values in an end-to-end fashion.

majority of the combinations are from NCI Almanac study, and it is also the largest individual study in the DrugComb's database. The sizes of the dose-response matrices differ across studies, and it even differs within the same study (i.e., NCI Almanac study has dose-response matrices with both 4×4 and 4×6).

2.3 Input Features

We use two types of features to train the model. The first set of features encodes the untreated cells' gene expression (GEX) profiles. The data were obtained from Iorio *et al.* [28] (accession number E-MTAB-3610 at ArrayExpress [29]). We downloaded the RMA normalized gene expression profiles from <https://www.cancerrxgene.org> (downloaded on Dec 2019). For each cell line k , we use the expression profiles of landmark genes, denoted as $c_k \in \mathbb{R}^{972}$. We obtain the list of landmark genes from Subramanian *et al.* [30]. The second set of features describes the drug pairs' chemical structure. To represent the drugs in the combination, drug chemical descriptor features $x \in \mathbb{R}^{541}$ are calculated using ChemoPy Python library [31]. The chemical descriptors and the number of features for each descriptor are provided in Table S1, which can be found on the Computer Society Digital Library at <http://doi.ieeecomputersociety.org/10.1109/TCBB.2021.3086702>.

2.4 The MatchMaker Architecture

MatchMaker is a deep neural network-based drug synergy prediction algorithm. The architecture is shown Fig. 2. The model inputs (i) the drug chemical descriptors of a pair of drugs and (ii) an untreated cell line gene expression profile as features to predict the Loewe score [21] of a drug pair. The model contains three neural subnetworks. First, there are two subnetworks that learn a representation of each drug conditioned on cell line gene expression of the given cell line. The output of these two subnetworks are then concatenated and input to a third subnetwork that predicts the final output. The individual components are detailed below.

2.4.1 Drug Specific Subnetworks (DSN)

For a drug pair, cell line trio, MatchMaker includes two Drug Specific Subnetworks, DSN_i and DSN_j for each drug i and j respectively. DSN_i inputs the i th drug's chemical representation (x_i) together with untreated gene expression profile of the k th cell line (c_k). Symmetrically, DSN_j inputs x_j and c_k . Both DSNs share the same architecture and apply three fully connected (FC) layers, followed by activation functions and dropout [32]. We use ReLU activation [33] at the first two FC layers and linear activation at the final layer of DSNs. Dropout probabilities are set to 0.2 and 0.5 at the first and second layers, respectively. No dropout is applied for the final layer. Each DSN produces a vector embedding for the corresponding drug conditioned on the cell line expression. We refer these embeddings as $x'_{i,k}$ and $x'_{j,k}$. The formulation of DSN can be summarized as follows:

$$\begin{aligned} O_{1,m,k} &= \text{Dropout}^{(p_1)}(\text{ReLU}(\text{FC}^{(n_1)}(\text{CAT}(x_m, c_k)))) \\ O_{2,m,k} &= \text{Dropout}^{(p_2)}(\text{ReLU}(\text{FC}^{(n_2)}(O_{1,m,k}))) \\ x'_{m,k} &= \text{Linear}(\text{FC}^{(n_3)}(O_{2,m,k})), \end{aligned} \quad (1)$$

where $x'_{m,k}$ represents the latent representation of the drug m with cell line k , which is the output of the DSN_m . $\text{FC}^{(*)}$ represents a fully connected layer with $*$ neurons and $\text{Dropout}^{(p)}$ represents a dropout layer with probability p . Linear and ReLU stand for linear and rectified linear unit activation functions, respectively. CAT concatenates given features vectors.

2.4.2 Synergy Prediction Subnetwork (SPN)

SPN takes the concatenated vector embeddings calculated by DSN_i and DSN_j subnetworks. SPN is a three-layered FC network. Similar to DSN, we use RELU activation in the first two FC layers and linear activation at the output layer. Dropout, with a drop probability of 0.5, is applied after the first ReLU activation layer. SPN produces the predicted Loewe score of the drug pair $\langle i, j \rangle$ on cell line k denoted as $\hat{y}_{i,j,k}$. The formulation of SPN can be summarized as follows:

$$\begin{aligned} O'_{3,i,j} &= \text{Dropout}^{(p_4)}(\text{ReLU}(\text{FC}^{(n_4)}(\text{CAT}(x'_{i,k}, x'_{j,k})))) \\ \hat{y}_{i,j,k} &= \text{Linear}(\text{FC}^{(1)}(\text{ReLU}(\text{FC}^{(n_5)}(O'_{3,i,j})))) \end{aligned} \quad (2)$$

We train MatchMaker in an end-to-end fashion using a weighted version of MSE as the loss function (Equation (3)). We modify the loss such that the examples that are more synergistic are weighted more. Our goal is to focus more on the examples that are highly synergistic because these are drug pair, cell line combinations with higher therapeutic potential. To achieve this, the weight of a trio in the loss calculation is set proportionally to its true Loewe score's distance from the minimum of all Loewe scores in the training data (Equation (4)). We use the weighted MSE just for training purposes, and evaluate validation and test folds with standard MSE

$$MSE_w = \frac{1}{N} \sum_{i,j,k \in D} w_{i,j,k} * (y_{i,j,k} - \hat{y}_{i,j,k})^2 \quad (3)$$

$$w_{i,j,k} = \ln(y_{i,j,k} - \min(\mathbf{y}) + e), \quad (4)$$

where N is the size of the training dataset D and $w_{i,j,k}$ is the loss the weight for a trio (drug i , drug j , cell line k). $\mathbf{y} \in \mathbb{R}^N$ denotes the vector of true labels in the training set, $y_{i,j,k}$ is the true Loewe score for the corresponding trio, $\hat{y}_{i,j,k}$ is the predicted score and e is the Euler's number.

2.5 Compared Methods

We compare the performance of MatchMaker with two available methods from the literature, which also exclusively use drug structure and the untreated gene expression profiles of the cell lines. DeepSynergy is a fully-connected multi-layered deep neural network based algorithm [18]. The second algorithm is TreeCombo, in which Janizek *et al.* [19] use XGBoost algorithm [20] to predict the Loewe score of drug pair $\langle i, j \rangle$ in cell line k . Both methods input a concatenated vector of the drug chemical structure, and cell line features: $\text{CAT}(x_i, x_j, c_k)$. Note that this is different from MatchMaker architecture, which first learns an embedding per drug on the cell line of interest and then concatenates the embeddings to learn synergy score prediction.

We also compare MatchMaker with two other standard machine learning methods: (i) Random Forest, an ensemble method based on decision trees, and (ii) ElasticNet, a linear regression method that combines L_1 and L_2 regularization. To be consistent with the previous two methods, Random Forest [34] and ElasticNet [35] are also trained with the same concatenated input $\text{CAT}(x_i, x_j, c_k)$. Finally, we calculate a baseline model that for a drug pair-cell line trio, $\langle i, j, k \rangle$, predicts the label as the mean of the synergy scores for all training examples that involve the cell line k : $\hat{y}_{i,j,k} = \frac{\sum_{\langle i,j \rangle \in D^k} y_{i,j,k}}{|D^k|}$ where D^k contains all experiments performed on cell line k .

3 RESULTS

3.1 Experimental Setup

For performance assessment, we split the dataset into train, validation, and test folds with ratios 60, 20, and 20 percent, respectively. To ensure a fair comparison, we stratify the

samples into folds using *leave-drug-combination-out approach* as described in Preuer *et al.* [18]. In this approach, a drug pair appear in only either one of the train, validation, or test sets. We tune the parameters on the validation set and final performances are reported on the test set. All algorithms are compared using the train, validation, and test splits.

In addition to the above-mentioned setting, we re-evaluate MatchMaker using 5-fold cross-validation (CV) to assess the robustness of the results to different data splits. The folds are rotated in each CV run; in each CV run, the three folds are used for training, one fold for validation and one for testing. This way, the CV procedure yields in 5 trained models and we obtain a single prediction per drug pair and cell line combination as each example appears only once in the test fold. We also repeat the CV procedure 10 times with different random fold splits. As a result, we obtain 10 predictions per drug pair and cell line combination.

We train the compared models on the same dataset that we use to build and evaluate MatchMaker. We evaluate various hyperparameter combinations to train DeepSynergy, such as different data normalization techniques $\in \{\tanh, \tanh_norm\}$, different number of layers $\in \{2, 3\}$ with different numbers of neurons $\in \{512, 1024, 2048, 4096\}$ (see Table S2, available in the online supplemental material). For the TreeCombo model, we set the *number of estimators* parameter to 1000 as also in [19]. We also tune the *maximum tree depth* $\in \{4, 6, 8, 10, 12\}$ and *learning rate* $\in \{0.05, 0.10, 0.15\}$ parameters as suggested by Janizek *et al.* (see Table S3, available in the online supplemental material). For the Random Forest model, we choose the best performing *Maximum tree depth* $\in \{4, 6, 8, 10, 12\}$ and *the number of trees* $\in \{10, 100, 500, 1000\}$ parameters (Table S4, available in the online supplemental material). Finally, for ElasticNet *L1 ratio* $\in \{0.20, 0.40, 0.60, 0.80\}$ and $\alpha \in \{0.001, 0.01, 0.1, 1.0, 10, 100\}$ parameters are considered (Table S5, available in the online supplemental material).

We train MatchMaker using early stopping mechanism [36]. That is, while minimizing the training loss, we monitor the validation loss and terminate the training if validation loss does not improve for p epochs ($p = 100$) and save the model that provided the best validation loss. We set the maximum number of epochs to 1000. We use the Adam optimizer [37] with the learning rate of 10^{-4} and batch size 128 to the loss function. We use Xavier initialization [38]. During training, a sample for a drug pair $\langle i, j \rangle$ is input to the network also in the reversed order, $\langle j, i \rangle$. This is to train each DSN with every drug and to avoid order bias. During testing, we obtain predictions for both $\langle i, j \rangle$, and $\langle j, i \rangle$ for all methods, and the final Loewe score is obtained by averaging these two predicted scores.

We train all the models on a SuperMicro SuperServer 4029GB-TRT with 2 Intel Xeon Golf 6140 Processors (2.3GHz, 24.75M cache) and 251GB RAM. MatchMaker and DeepSynergy are trained on 1 NVIDIA TITAN RTX GPU (24GB, 384Bit). The training times were approximately 5 hours for MatchMaker and 4 hours for DeepSynergy. We use Keras [39] with TensorFlow [40] backend for MatchMaker and DeepSynergy. We use scikit-learn [41] implementations for Random Forest and ElasticNet and the Python package of XGBoost to implement TreeCombo [20]. We train TreeCombo, ElasticNet and Random Forest on CPU, and their

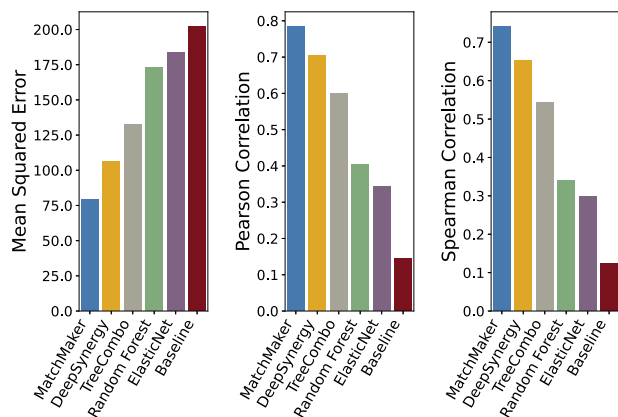


Fig. 3. Performance comparison of models based on MSE, Pearson and Spearman correlation coefficients. MatchMaker achieved the best performances with 79.49 MSE, 0.79 Pearson and 0.74 Spearman's correlation coefficients.

training times were approximately 1 hour, 30 mins and 5 hours, respectively.

3.2 Performance Comparison

3.2.1 Regression Performance

We compare the models' predictive performances on the held out drug combinations using three performance metrics: (i) MSE, (ii) Pearson, and (iii) Spearman rank correlation coefficients between the actual Loewe scores and the predicted Loewe scores. Fig. 3 summarizes the model performances for different algorithms and for different performance metrics. MatchMaker achieves the lowest MSE value, 79.49, while DeepSynergy's is the next best performer with an MSE value of 105.9. TreeCombo's predictions lead to 132.7 MSE. This corresponds to a 33 percent improvement over DeepSynergy and a 67 percent improvement over TreeCombo in terms of MSE. Of all the methods, Random Forest and ElasticNet perform worst and yield to the highest MSE values: 173.2 and 183.4, respectively. Note that the baseline model leads to 202.5 MSE.

When models are evaluated in terms of the Pearson and Spearman rank correlation coefficients, a similar trend is observed. MatchMaker achieves the best performance in terms of Pearson, 0.79, and 0.74 Spearman correlation coefficients. Overall, the rankings of the models are the same across the three performance metrics and MatchMaker is consistently the best performing one among the five algorithms compared.

Next, we investigate if cell line gene expression data bring predictive power. MatchMaker trained in the same experimental settings but without the cell line gene expression features leads to an MSE of 131.9, Pearson correlation of 0.60, and Spearman correlation of 0.53. This clearly shows that cell line gene expression data is an important ingredient for the model, resulting in a substantial performance gain.

3.2.2 Classification Performance

We further assess the trained regression model's ability to distinguish synergistic drug combinations and mimic a classification task for evaluation as was previously done in the literature [18]. That is, we binarize the actual synergy scores ($y \in \mathbb{R}$) such that those larger than a positive Loewe score

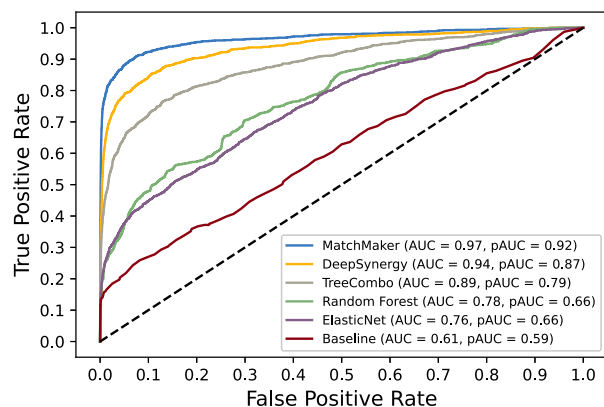


Fig. 4. Receiver operating characteristics (ROC) curves for all methods in the classification task. Synergistic combinations with a Loewe score higher than 10 are classified as positive samples and those below -10 are classified as negative. Corresponding AUC values and partial AUC (pAUC) values up to 0.1 FPR are also noted.

threshold, t , constitute the positive class (synergistic) pairs ($\bar{y} = 1$). In contrast, those less than $-t$ constitute the negative class ($\bar{y} = 0$). We decide threshold t based on the user guide of the SynergyFinder tool [42], which is also developed by the DrugComb team [22]. They state therein that there is no specific threshold for determining synergy, and the scores close to zero do not correspond to confident synergistic or antagonistic predictions. They suggest that the pairs with synergy scores less than -10 are likely to be antagonistic while the pairs with synergy scores greater than 10 are likely to be synergistic. The ones that are in between -10 and 10 can be classified as additive. Therefore, we follow the same reasoning and use $(-10, 10)$ as thresholds for the positive and negative classes. We additionally experiment with lower and higher thresholds, where t is 0 and 20. We evaluate the predictive performance of the models on this binary classification task by the area under the receiver operating characteristics (ROC) curve. Fig. 4 shows ROC curves for the different models when t is set to 10. Based on AUC, partial AUC (false positive rate up to 0.1; pAUC) and area under the precision recall curve (AUPR), MatchMaker outperforms the other compared methods by a large margin (see Table 3).

The same classification success holds when t is varied (see Figs. S2 and S3; and Tables S6 and S7, available in the online supplemental material). When t is set to 20 instead of 10, performances for all models increases as distinguishing very synergistic and very antagonistic examples is a relatively easier classification task. When t is set to 0, we are testing whether the model is able to differentiate the positive and negative Loewe scores (Fig. S2 and Table S6, available in the online supplemental material). This is a more difficult task. Also note that, in this case, the definitions of the true positive and true negative are clouded by noise, as a Loewe score just barely above zero is not a very strong indicator of synergy. Even though the performances for all the models decrease when $t = 0$, MatchMaker still outperforms other methods in terms of the classification performance metrics.

The ultimate goal of in-silico synergy prediction models is to prioritize the candidates to be validated experimentally. Therefore, it is relatively more important to achieve

TABLE 3
Performance Comparison of the Methods in the Classification Task

Model	AUC	pAUC	AUPR
MatchMaker	0.97	0.92	0.85
DeepSynergy	0.94	0.87	0.74
TreeCombo	0.89	0.79	0.61
Random Forest	0.78	0.66	0.37
ElasticNet	0.76	0.66	0.36
Baseline	0.61	0.59	0.23

Area under ROC curve (AUC), partial AUC calculated up to false positive rate 0.1 (pAUC), and area under precision-recall curve (AUPR) are shown for all methods considered. Experiments with a Loewe score higher than 10 were classified as positive (synergistic) samples and those below -10 are classified as negative (antagonistic) samples.

higher performance at the top candidates with the highest predicted synergy scores. We also compare the models in terms of the pAUC score which considers AUC in a restricted false positive rate (FPR) range. When we compare the methods using pAUC up to 10 percent FPR, we observe that MatchMaker pAUC improvement is even larger than the pAUC improvement of the second best method. MatchMaker achieves 0.92 pAUC score, whereas DeepSynergy achieves 0.85 when t is set to 10. The same trend holds when t is set to 0 and 20 as well (Figs. S2 and S3; and Tables S6 and S7, available in the online supplemental material).

3.3 Confidence on Predictive Performance of MatchMaker

In addition to the results in Section 3.2, we assess the confidence of the MatchMaker’s performance by repeating the 5-fold CV procedure 10 times with random splits at each time as described in Section 3.1. In Fig. 5 the average ROC curve is shown with a solid line and the shaded area represents the error bars with one standard deviation. This result indicates that the predictive performance do not vary across different random splits of the data.

We also check the regression performances in the same CV setting. MatchMaker achieves 82.3 average MSE with a standard deviation (std.) of 2.6 while average Pearson and Spearman correlations are 0.78 (std. 0.009) and 0.75 (std. 0.009) respectively. The CV results are similar to our results presented in Section 3.2. The small standard deviations suggest that

MatchMaker’s performance is robust against changes in the training set ups (e.g., different train folds and initial weights).

3.4 MatchMaker Performance Across Cell Lines and Drug Pairs

To understand the extent of variability of the model performance across cell lines, we first calculate the cell line specific performance metrics. That is, we measure the performance for all experiments performed on a cell line k . Fig. 6 shows that there are few cell lines where the model performs poorly. The cell lines with very high MSE values are T98G, L-1236 and NCIH2122. We inspect the reason that could explain this performance difference. Yet, we do not see a clear association to the number of training examples available (1036, 1391, and 290, respectively) or their tissue types (brain, haematopoietic/lymphoid, and lung, respectively). Interestingly, when we analyze the consistently-predicted false positives (i.e., novel predictions), many experiments on T98G cell line are predicted to be false positive with a large MSE gap (see Section 3.5). Thus, one possibility is that the actual labels for these cell lines could be noisy and there is a systematic error on these cell lines’ experiments. There are also a few cell lines for which the model performs remarkably well. For the top 6 performing cell lines, the MSE is below 30. The best performing three cell lines are RD, A2780 and A427. When we inspect the correlation metrics, we observe that for 75 of the 81 cell lines, the Pearson correlation coefficient is above 0.7, indicating that for most of the cell lines MatchMaker performance is good. Finally, we observe for 71 of the 81 cell lines, the Spearman rank correlation coefficient is above 0.7. We provide these cell line specific evaluations in Fig. S1 and Table S8, available in the online supplemental material.

Next, we evaluate the drug-pair specific predictive performance of MatchMaker . We consider the drug pairs that appear with more than one cell line in the drug pair-cell line combinations in the test data. Table 4 displays the top 10 drug pairs that yield the minimum MSEs. Over the 1082 unique drug pairs, we observe that 74 percent of them are below the overall MSE of 79.49. In terms of the correlation, we observe 46 and 35 percent of drug pairs are above 0.7 for Pearson and Spearman correlation coefficients, respectively. Thus, in terms of correlation metrics, the performance with respect to drug pairs vary more compared to performance with with respect to the cell lines. We see that all but one top-10 drug pairs have

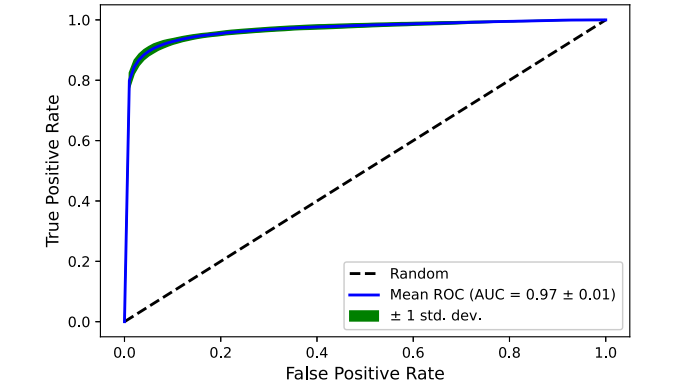


Fig. 5. Area Under Receiver operating characteristics (ROC) curve for MatchMaker and confidence interval in classification task where the threshold is 10.

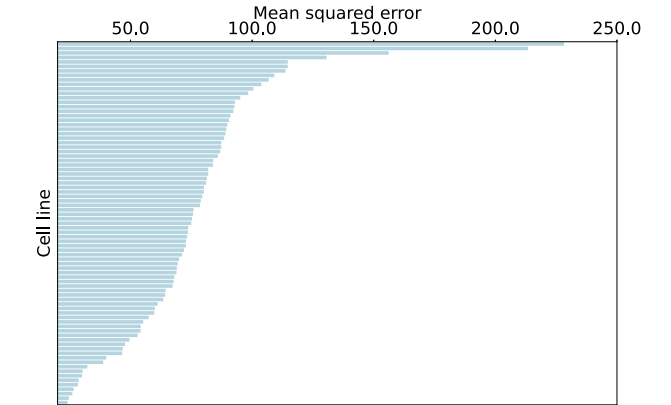


Fig. 6. Cell line specific MSE performance of MatchMaker is shown.

TABLE 4
The Top Predictive Drug Pairs Based on MSE

Drug1	Drug2	MSE	Pair count
LENALIDOMIDE	IMIQUIMOD	4.93	54
BENDAMUSTINE HYDROCHLORIDE	PROCARBAZINE HYDROCHLORIDE	5.26	52
CELECOXIB	ERLOTINIB HYDROCHLORIDE	5.52	52
TENIPOSIDE	BORTEZOMIB	5.94	54
DEXRAZOXANE	PROCARBAZINE HYDROCHLORIDE	6.04	54
LOMUSTINE	ANASTROZOLE	6.07	53
LOMUSTINE	FULVESTRANT	6.13	54
BEZ-235	SN-38	6.37	27
VISMODEGIB	1-(5-DEOXYPENTOFURANOSYL)-5-FLUORO-4- [[PENTYLOXY)CARBONYL]AMINO}PYRIMIDIN-2(1H)-ONE	6.46	54
MITOXANTRONE	NSC733504	6.61	54

over 50 experiments in the test dataset. The average number of experiments (i.e., number of cell lines investigated) per drug pair is 25. This shows that the model performance overall benefits from large number of training samples.

3.5 Novel Predictions

The ground truth labels for experiments in DrugComb could be noisy or could have systematic bias. Here, we check the drug-cell line combinations that are consistently predicted as false positives by MatchMaker. That is, the model predicts a particular drug pair in a cell line to interact in a synergistic fashion, where indeed it is labeled as antagonistic in the ground truth.

We find the trios that are consistently predicted as false positive in all 10 random initialized MatchMaker models. Note that they are in the test set 10 times despite we employ a 5-fold cross validation scheme as described in Section 3.1. Out of the 286,421 combinations, we find 2,540 drug pair-cell line combinations that are consistently marked as false positives. To get the ones that are more confident, we retain the combinations with actual Loewe score less than -10 and predicted Loewe score larger than 10. This leaves us with 25 combinations, which are provided in Table 5. We present the whole list of 2,540 drug pairs in Table S9, available in the online supplemental material.

For these 25 trios, we also inspect drug combination sensitivity scores (CSS), and the other synergy measures: Bliss independence (BLISS) [43], Highest single agent (HSA) [44] and Zero interaction potency (ZIP) [45] and S synergy score [46]. For the two of the 25 trios, all of these measures indicate a synergistic interaction, while for the rest of the trios, there is a varying degree of support. We list this extra information in Table 5.

The first two trios in Table 5 are: <VEMURAFENIB, PLICAMYCIN> and <VEMURAFENIB, ACTINOMYCIN D> tested on RPMI-8226 cell line (haematopoietic/lymphoid tissue). For these pairs, (i) MatchMaker predicts a synergistic relationship, (ii) Loewe score indicates an antagonistic interaction, and finally, (iii) all other measures are in favor of synergism. For one of these pairs we indeed find evidence in the literature: VEMURAFENIB and ACTINOMYCIN D interacts and VEMURAFENIB can increase the serum concentration of ACTINOMYCIN D [47].

The following six combinations in Table 5 are supported by least three out of four synergy measures and by a

positive CSS score. Among them, the PENFLURIDOL and TEMOZOLOMIDE (TMZ) combination has been shown to prolong the survival by (i) decreasing the tumor growth [48], and (ii) increasing the apoptosis in Glioblastoma (GBM) tumors [49]. The drug pair reduces the MGMT and GLI1 expression in T98G cell line along with several other GBM cell lines [49]. MatchMaker is also in disagreement with the actual truth Loewe scores in combinations of TMZ with (i) TG100-115, (ii) CHLORPROPAMIDE, and (iii) LY2109761. We find evidence in the literature supporting the effectiveness of these drug combinations. Li *et al.* [50] state that TMZ enhances the effect of TG100-115 on GBM cells and improves survival in mice [51]. Zhang *et al.* [52] show that the combination of LY2109761 and TMZ suppresses the tumor growth on T98G cell line. Finally, we find that CHLORPROPAMIDE can increase the serum concentration of TMZ [47].

Table 5 shows that for two drugs the difference between the predicted and the actual synergy scores are off by a large margin for many cases: (i) PRALATREXATE and (ii) TMZ. Moreover, these experiments are frequently performed on the T98G cell line. T98G is one of the cell lines on which MatchMaker performs the worst on average (see Fig. 6). As discussed in Section 3.4, systematic error on this set of T98G experiments could be an explanation for these contradictory predictions.

3.6 Evaluation on Other Datasets

We further evaluate MatchMaker on two other datasets: O'Neil dataset [23] and NCI ALMANAC dataset [25] to assess if the observed performance improvement over the state of the art methods holds across different datasets. The O'Neil dataset includes 23,062 samples that covers 39 cell lines and 38 distinct drugs. We remove 9 of the 39 cell lines because these cell lines (gene expression features) are not included in the untreated cell line gene expressions of E-MTAB-3610 [28]. After this filtering, a total of 18,346 samples of 30 cell lines and 38 unique drugs are covered in the datasets. We obtain the Loewe scores for these examples from Preuer *et al.* [18] for the O'Neil dataset.

The NCI ALMANAC dataset has 311,604 combinations with 108 drugs and 60 cell lines. We eliminate 5 of 60 cell lines since they are not found in the E-MTAB-3610 cohort [28]. We also further eliminate 9 drugs because two or more drugs map to same PubChem CID or drug's structure

TABLE 5
The List of Confident False Positives Which are Designated as Novel Synergistic Drug Pair Predictions

Drug1	Drug2	Cell line	Actual Loewe	Predicted Loewe	CSS	S	ZIP	BLISS	HSA	Support
VEMURAFENIB	PLICAMYCIN	RPMI-8226	-47.72	18.45	30.78	20.37	3.76	8.00	0.51	*****
VEMURAFENIB	ACTINOMYCIN D	RPMI-8226	-24.6	24.67	58.83	22.70	10.29	12.72	9.72	*****
HYDROXYUREA	PRALATREXATE	K-562	-37.8	24.44	10.37	5.02	6.22	4.32	-5.08	*****
VINBLASTINE SULFATE	PRALATREXATE	K-562	-10.84	35.41	34.80	18.08	-2.59	5.02	1.34	*****
WNT-C59 (C59)	TEMOZOLOMIDE	T98G	-14.22	18.27	3.17	9.00	4.96	4.96	-14.22	*****
SESAMIN	TEMOZOLOMIDE	T98G	-10.65	19.62	6.75	8.44	5.12	5.12	-10.65	*****
PENFLURIDOL	TEMOZOLOMIDE	T98G	-12.97	15.54	4.43	18.75	13.24	13.24	-12.97	*****
RUPADADINE FUMARATE	TEMOZOLOMIDE	T98G	-11.17	15.58	6.23	5.18	2.33	2.33	-11.17	*****
SORAFENIB	PRALATREXATE	K-562	-86.9	24.64	29.61	22.02	-0.63	3.85	-5.38	***
HYDROXYUREA	PRALATREXATE	786-0	-36.42	18.37	33.89	24.84	-0.69	0.51	-1.10	***
VINBLASTINE SULFATE	PRALATREXATE	786-0	-11.96	39.38	7.64	7.53	-0.94	0.52	-1.16	***
ANASTROZOLE	PRALATREXATE	786-0	-12.23	25.00	7.21	7.23	-1.73	0.61	-0.84	***
BLEOMYCIN	PRALATREXATE	PC-3	-22.23	19.17	36.72	16.26	-4.39	-1.54	-0.59	**
VINBLASTINE SULFATE	PRALATREXATE	PC-3	-48.9	38.30	17.56	-7.60	-0.47	-2.53	-3.07	*
MITOTANE	PRALATREXATE	PC-3	-52.63	24.07	14.19	-10.75	-0.65	-4.63	-6.65	*
BENDAMUSTINE HYDROCHLORIDE	PRALATREXATE	HCT116	-24.94	37.18	0.62	-0.80	-0.77	-0.73	-7.08	*
JTC-801	TEMOZOLOMIDE	T98G	-30.02	14.30	54.64	-47.33	-32.62	-32.62	-29.94	*
TG100-115	TEMOZOLOMIDE	T98G	-13.09	24.01	18.05	-30.46	-25.04	-25.04	-13.06	*
CHLORPROPAMIDE	TEMOZOLOMIDE	T98G	-12.37	21.22	5.02	-13.91	-13.64	-13.64	-12.37	*
DOXOFYLLINE	TEMOZOLOMIDE	T98G	-16	16.74	13.49	-33.37	-28.24	-28.24	-15.97	*
GDC-0879	TEMOZOLOMIDE	T98G	-15.25	15.98	2.14	-9.92	-10.85	-10.85	-15.25	*
AZD2014	TEMOZOLOMIDE	T98G	-13.29	17.66	59.96	-30.62	-17.89	-17.89	-13.23	*
LY2109761	TEMOZOLOMIDE	T98G	-10.39	20.00	7.00	-3.83	-4.97	-4.97	-10.39	*
WZ811	TEMOZOLOMIDE	T98G	-10.82	12.53	6.57	-20.69	-18.97	-18.97	-10.82	*
FULVESTRANT	PRALATREXATE	DU-145	-12.35	27.61	-2.17	-8.59	-1.42	-4.0	-5.06	*

The pairs that have actual Loewe score less than -10 and have MatchMaker -predicted Loewe score larger than +10 in all 10 random initializations of the model are listed. The subsequent five columns of the table list alternative drug sensitivity (CSS) and drug combinations sensitivity scores (S, ZIP, BLISS, HSA). The last four sensitivity scores are obtained from DrugComb database. The last column summarizes the support of these alternative measures. That is, the number of stars indicates the number of alternative measures that also indicate a synergistic interaction.

TABLE 6
O’Neil and NCI ALMANAC Datasets Details

	O’Neil Dataset			NCI ALMANAC		
	Combinations	Drugs	Cell lines	Combinations	Drugs	Cell lines
Initial dataset	23,062	38	39	311,604	108	60
After feature filtering	18,346	38	30	256,711	99	55

information is not available at PubChem database [27]. The final processed NCI ALMANAC dataset includes 256,711 drug pair and cell line combinations of 55 cell lines and 99 drugs. The synergy scores of NCI ALMANAC is measured by ComboScore instead of Loewe score. Holbeck *et al.* [25] describe ComboScore as a metric that measures the interaction between two drugs in a given cell line. ComboScore is the sum of differences between observed and expected growth fractions across all dose combinations. The expected growth of two drugs A and B is calculated as follows:

$$E^{A_pB_q} = \begin{cases} \min(Y^{A_p}, Y^{B_q}) & \text{if } Y^{A_p} \leq 0 \text{ or } Y^{B_q} \leq 0 \\ \frac{Y^{A_p} * Y^{B_q}}{100} & \text{otherwise} \end{cases},$$

(5)

where Y^{A_p} and Y^{B_q} are the observed growth fractions of drugs A and B with concentrations p and q . We summarize the number of samples, drugs, and cell lines at Table 6.

In both datasets, we perform the same hyperparamater tuning procedure as described in Section 3.1 for all models. We compare models’ performances in terms of MSE, Pearson and Spearman correlations. The lowest MSE value on the

O’Neil dataset, 267.9, is achieved with MatchMaker while TreeCombo achieves the second-best MSE value of 283.1, and DeepSynergy leads to 290.0 MSE. On the NCI ALMANAC dataset, MatchMaker achieves a more considerable performance gap with MSE of 3014.5 while TreeCombo and DeepSynergy lead to 3651.6 and 3881.6 MSE values. In both datasets, we observe similar results for correlation coefficients. MatchMaker leads to 0.59 Pearson and 0.55 Spearman correlations for the NCI ALMANAC dataset while they are 0.69 in the O’Neil dataset. Table 7 summarizes models’ performances in O’Neil and NCI ALMANAC datasets. The performances of all models decrease in O’Neil and NCI ALMANAC datasets. This decrease could be due to different properties of the datasets. One of them is the synergy metrics being different. The NCI ALMANAC data uses a metric called ComboScore; its calculation is different from the Loewe score and its range (min and max synergy scores) is much wider than the Loewe scores in DrugComb. The ComboScore of NCI ALMANAC ranges in (−1318.7, 1144.3) while Loewe score of DrugComb ranges in (−116.8, 86.7). Since both the synergy measures and their ranges are different, it influences both MSE and correlation-based evaluation metrics. For experiments with the O’Neil dataset, the

TABLE 7
Performance Comparison of the Methods in the O'Neil and NCI ALMANAC Datasets Based on MSE, Pearson and Spearman Correlation Coefficients

	O'Neil dataset			NCI ALMANAC dataset		
	MSE	Pearson	Spearman	MSE	Pearson	Spearman
MatchMaker	267.9	0.69	0.69	3014.5	0.59	0.55
DeepSynergy	290.0	0.66	0.67	3881.6	0.40	0.36
TreeCombo	283.1	0.67	0.67	3651.3	0.47	0.47
Random Forest	342.4	0.57	0.58	3932.3	0.38	0.39
ElasticNet	305.3	0.62	0.63	4176.5	0.30	0.33

number of samples is lower compared to the number of samples in the DrugComb dataset; the O'Neil dataset includes 18,346 samples while the DrugComb dataset includes 286,421 samples. Since we have 15 times more samples in DrugComb and the machine learning models usually perform better with larger training set size, this might have contributed to the better performance of the models with the DrugComb data. Although the number of samples and the synergy score metrics are different, MatchMaker still achieved the best performances in both datasets.

4 DISCUSSION

In this study, for the first time, we made use of the DrugComb dataset [22], which contains close to half a million drug combination experiments. This is an invaluable source for training complex models. We observe that the large dataset size and a new architecture that learns drug specific weights enable MatchMaker to improve the prediction power of the state-of-the-art approaches substantially. We investigate the benefit of having more training examples for a drug. We observe that drugs that have more than 100 training samples in the training data have an average MSE of 79.43, Pearson correlation coefficient of 0.79, and Spearman correlation coefficient of 0.79. This is in contrast to the drugs with less than 100 training samples which have MSE of 161.69, Pearson correlation coefficient of 0.64, and Spearman correlation coefficient of 0.68. We foresee that with the growing experimental dataset sizes will enable more complex architectures to be trained and pave the way to more precise drug synergy predictions. Although we have a very large dataset for training today, the number of possible combinations to exhaustively validate experimentally is still intractable. This means accurate and efficient computational prioritization of novel synergistic drug pair candidates will keep being an important research area.

In MatchMaker, we use only the drug chemical structure as the primary feature, which has been used in parallel tasks such as drug target identification [53] or drug side effect prediction [54], [55]. We also use the cell line specific gene expression profile to capture the context of the experiment. These features are selected on the basis of their general availability to ensure the model's broad applicability to different drug combinations. The model presented here can be enhanced by integrating other rich biological information available on drugs and their targets. It has been previously shown that network-based approaches also

provide powerful predictions [56]. Pathways and protein-interaction networks could be leveraged. However, when drug targets are not one of the nodes of these networks, it reduces the size of the training data that can be used. Despite this limitation, using this information without sacrificing a large amount of training data is a future work to be explored. Other than drug side effects, drug categorizations and drug-disease relationships can be further utilized to improve the performance of MatchMaker.

Loewe score was our choice of target label due to its widespread use in the literature for this task. However, there are other synergy measures that differ in the reference model of non-interaction such as BLISS, HSA, ZIP and S synergy score. It is straightforward to modify our framework to work with these alternative measures of synergy. The model can also be enhanced to predict all measures simultaneously in a multi-task learning setting, which will be explored in future work.

During the preprocessing of the drug combination dataset, we filtered out drug pairs for which the replicate experiments exhibit large disagreement on the biological activity of the drug pair. However, there were no replicate experiments for a large portion of the dataset (81 percent). This could be a limitation of the dataset, which might have introduced label noise in the training and evaluation processes. Another limitation is that the DrugComb dataset considers only cancer cell lines. Although the framework is applicable to other conditions and diseases, the model trained is inevitably biased towards cancer. The model still could serve as a pretrained model and be fine-tuned for other cell lines and drugs for other diseases.

5 CONCLUSION

Combination therapy is of great interest in drug development due to improved efficacy and reduced side effects. In this work, we present MatchMaker, a deep learning-based model that utilizes drug chemical structure and cell line specific expression patterns to predict if two drugs work synergistically. We trained this model on the largest experimental dataset released to date. Our results demonstrate that MatchMaker outperforms state-of-the-art approaches in terms of predicting the Loewe score. By pruning the otherwise unmanageable search space, MatchMaker provides a useful tool for prescreening and prioritizing the candidate drug pairs in silico.

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